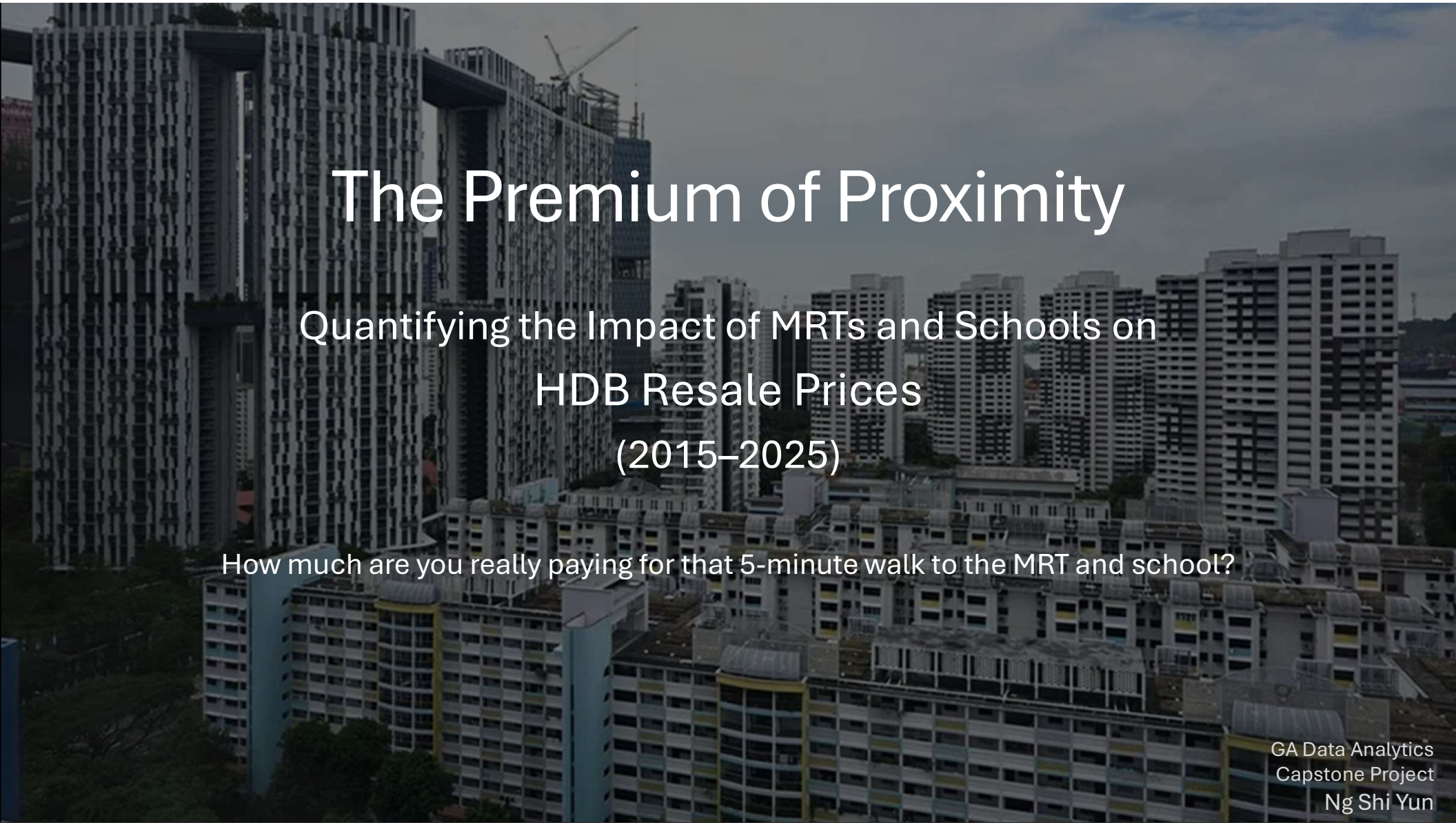


The Premium of Proximity

Quantifying the Impact of MRTs and Schools on HDB Resale Prices (2015–2025)

How much are you really paying for that 5-minute walk to the MRT and school?

GA Data Analytics
Capstone Project
Ng Shi Yun

Difficulty in pricing HDB resale flats

HDB resale prices vary by location, flat type, size, and age

Market conditions change over time

Agents often rely on experience or recent transactions

A data-driven reference can support better pricing decisions

Turning Addresses into Insights

The Source:

10 years of HDB Resale Transactions (Data.gov.sg).

Geospatial inputs:

Using Singapore's official OneMap API to calculate the exact walking distance from every HDB block to every MRT, top-tier school (GEP/SAP) and amenities.

Data sources

- HDB Dataset
- MOE school dataset
- Top primary school reference
- How Primary 1 registration balloting works
- MRT list from LTA's Public Transport page 2
- Bus stop
- Hawker Centers
- Parks
- Shopping malls
- CPI

Data Overview

Source: data.gov.sg – HDB - Resale Flat Prices
Period: 2015–2025 (10 years)
Full set: 260,019
Training set: 208,015
Test set: 52,004
Target variable: **Resale price (SGD)**

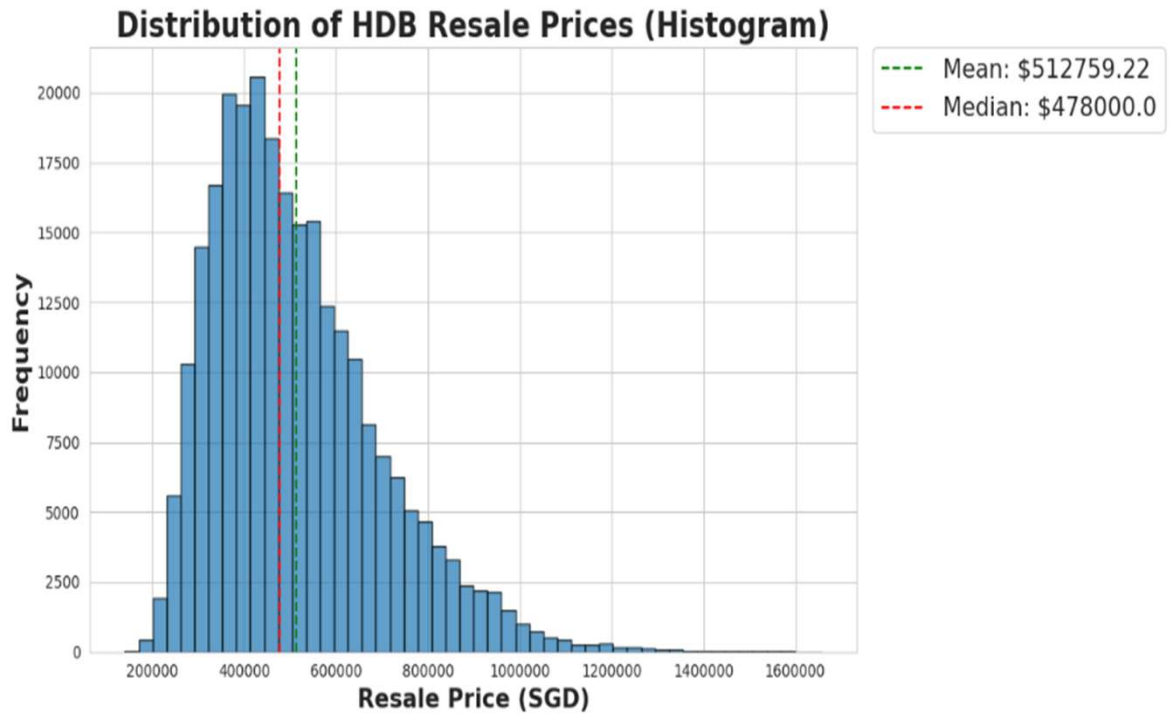
Feature categories:

1. Location (town, latitude/longitude)
2. Flat characteristics (type, size, model, storey, Hdb age)
3. Accessibility (MRT, mall, hawker distances)
4. Time (transaction year, lease commencement date)
5. Economic (CPI)

Pricing decisions must account for skewed distributions

Not Evenly Distributed

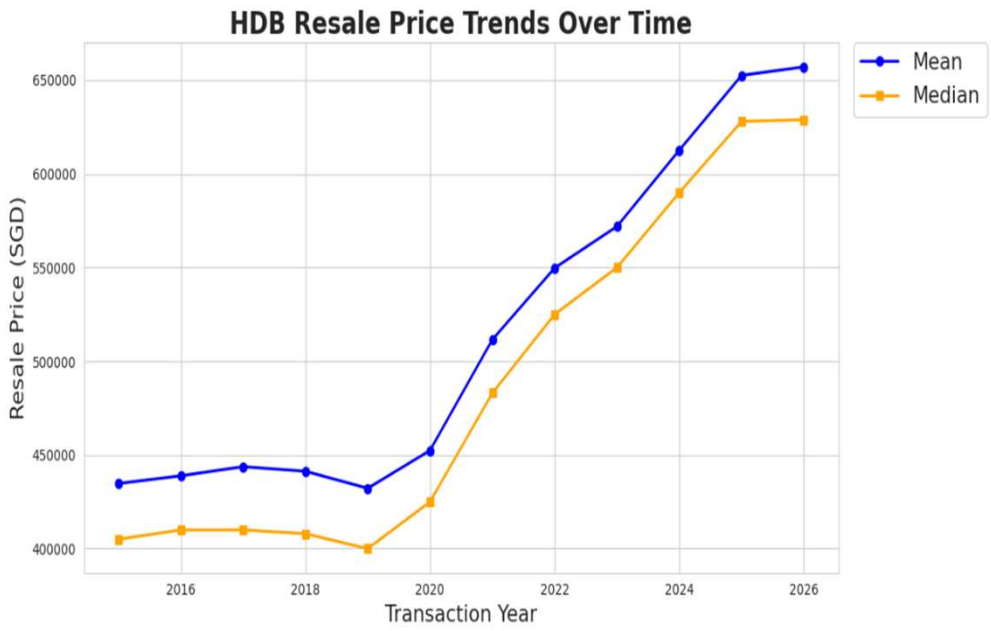
- Majority of transactions cluster in the mid-price range
- A smaller number of high-value transactions pull averages upward
- Median price is a more reliable benchmark than mean



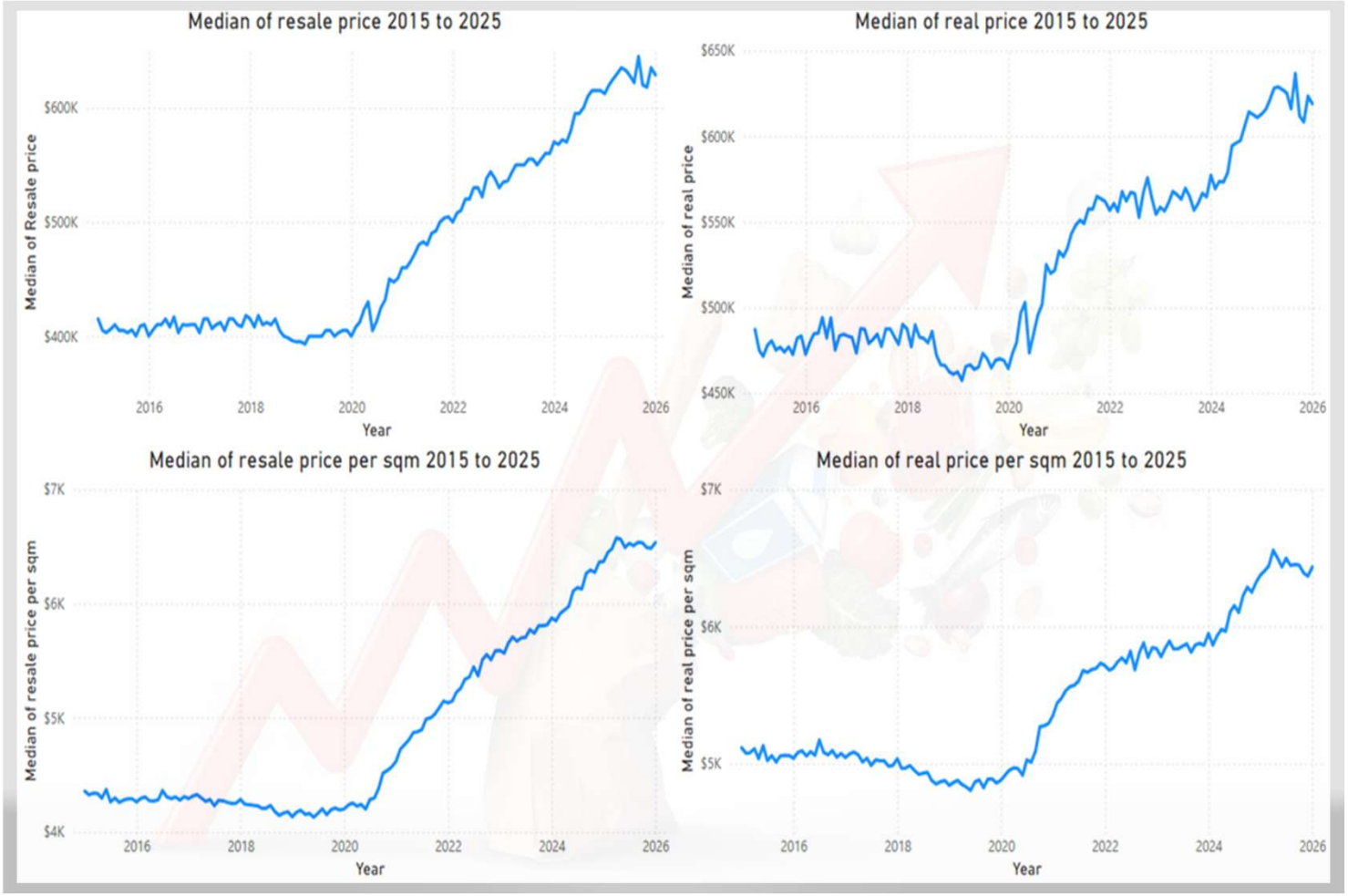
Increasing Price Trends Over Time

Resale Prices Reflect Market Cycles

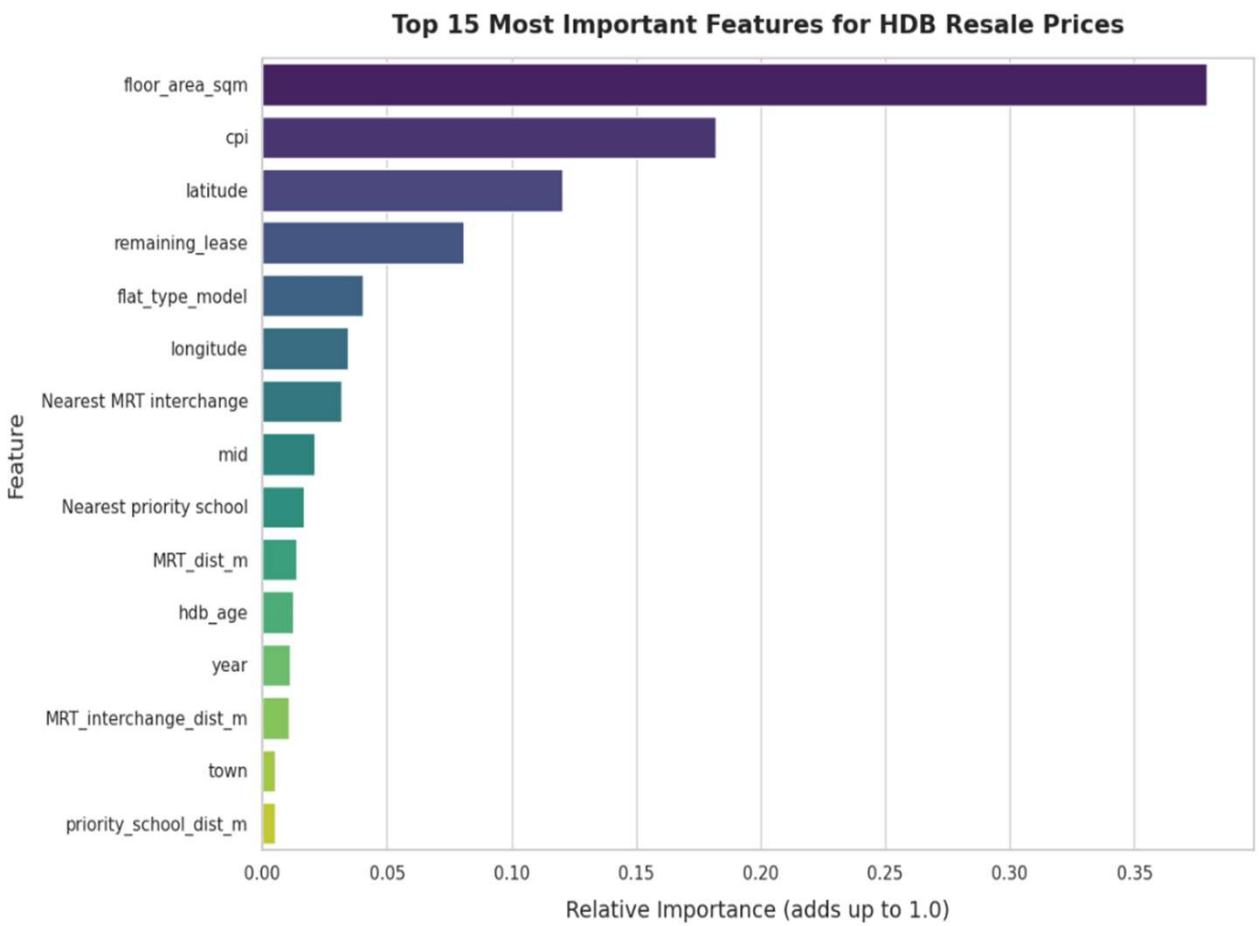
- Prices change over time rather than remaining static
- Periods of growth and stabilization are visible in transaction data
- Mean prices remain consistently higher than median prices, indicating right-skewed distribution



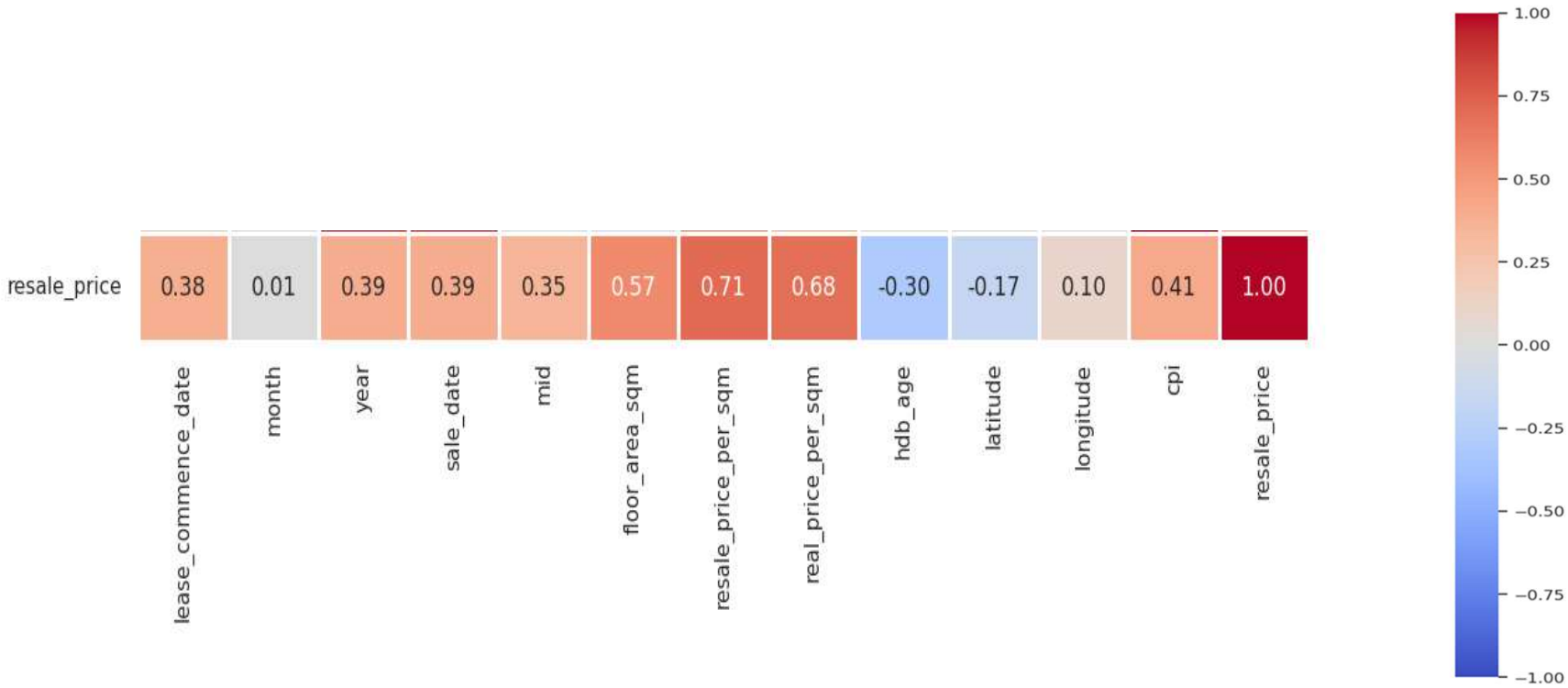
After adjustment for Inflation



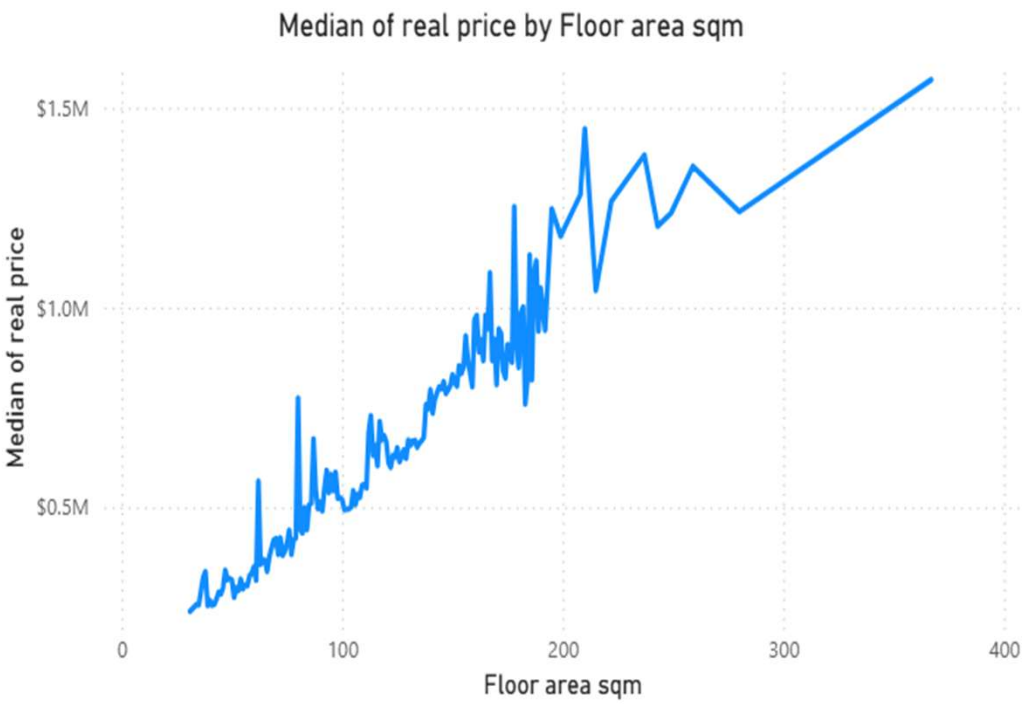
🏠 Price Difference drivers

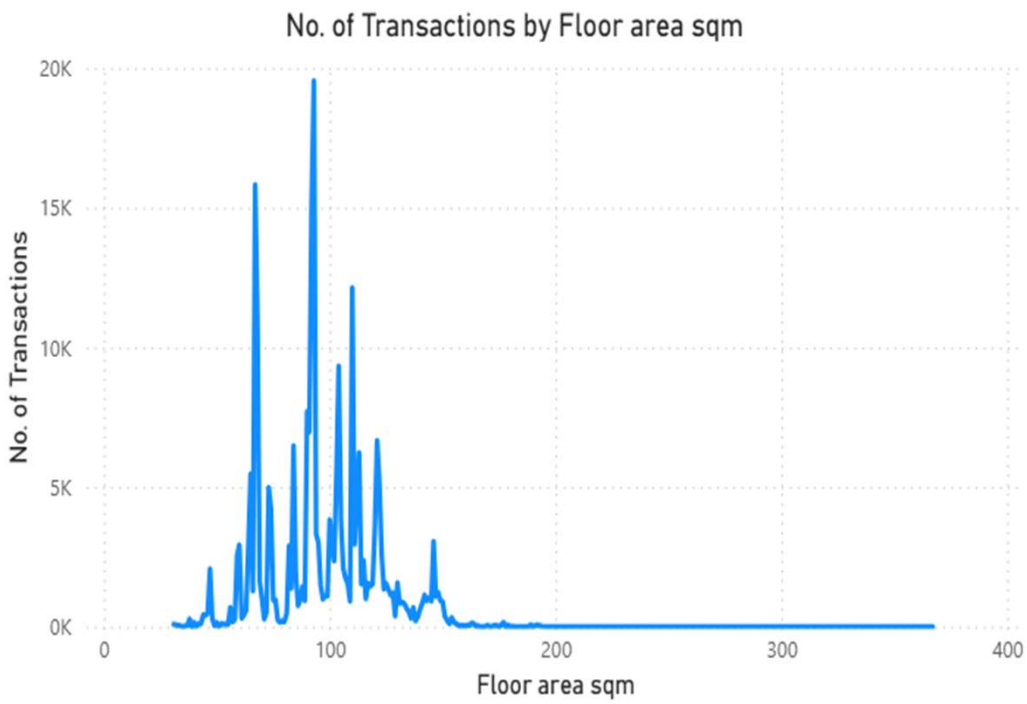


Price Difference drivers

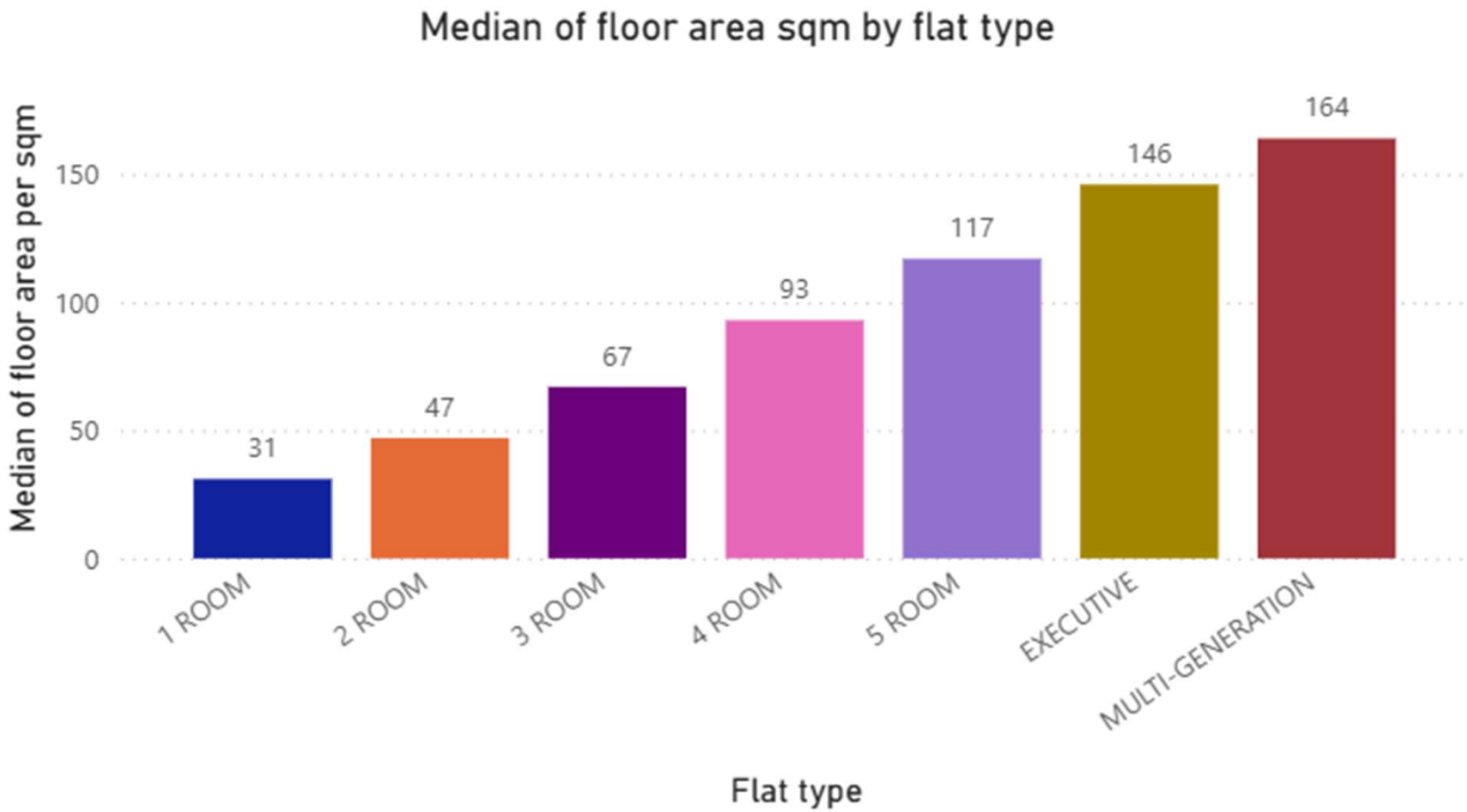


Bigger house, Higher Price

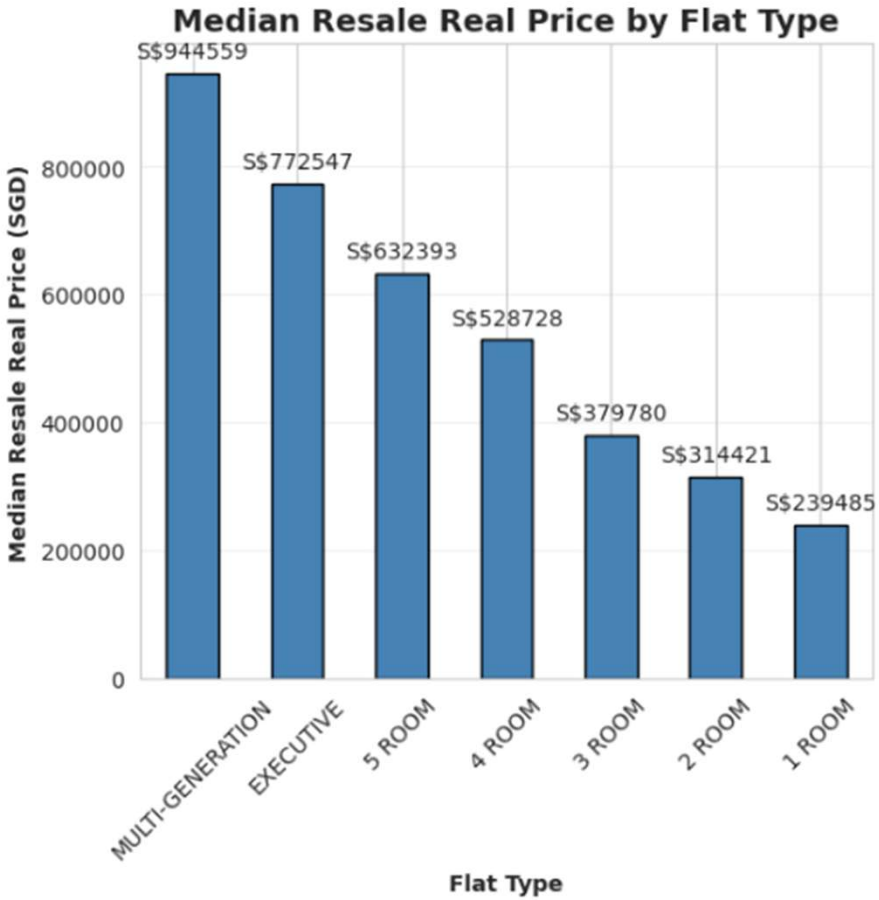




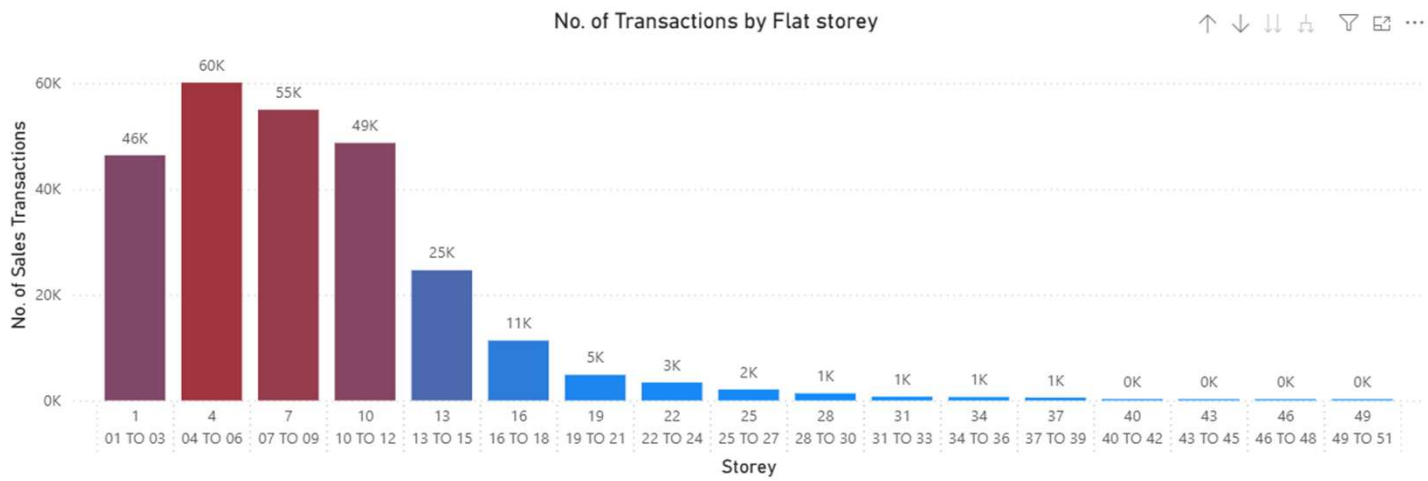
Flat types that tend to be larger

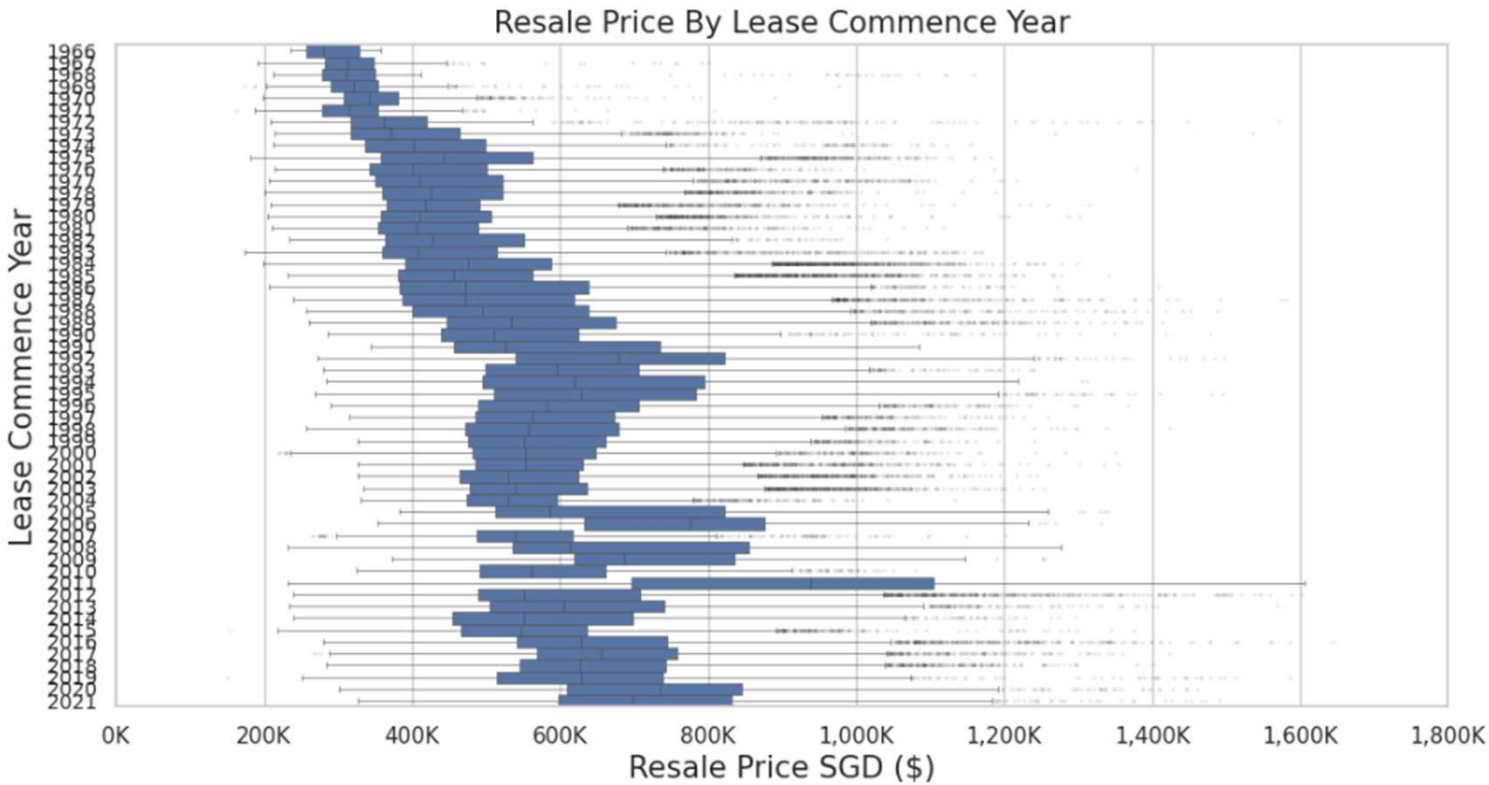


Price Distribution of Different flat types



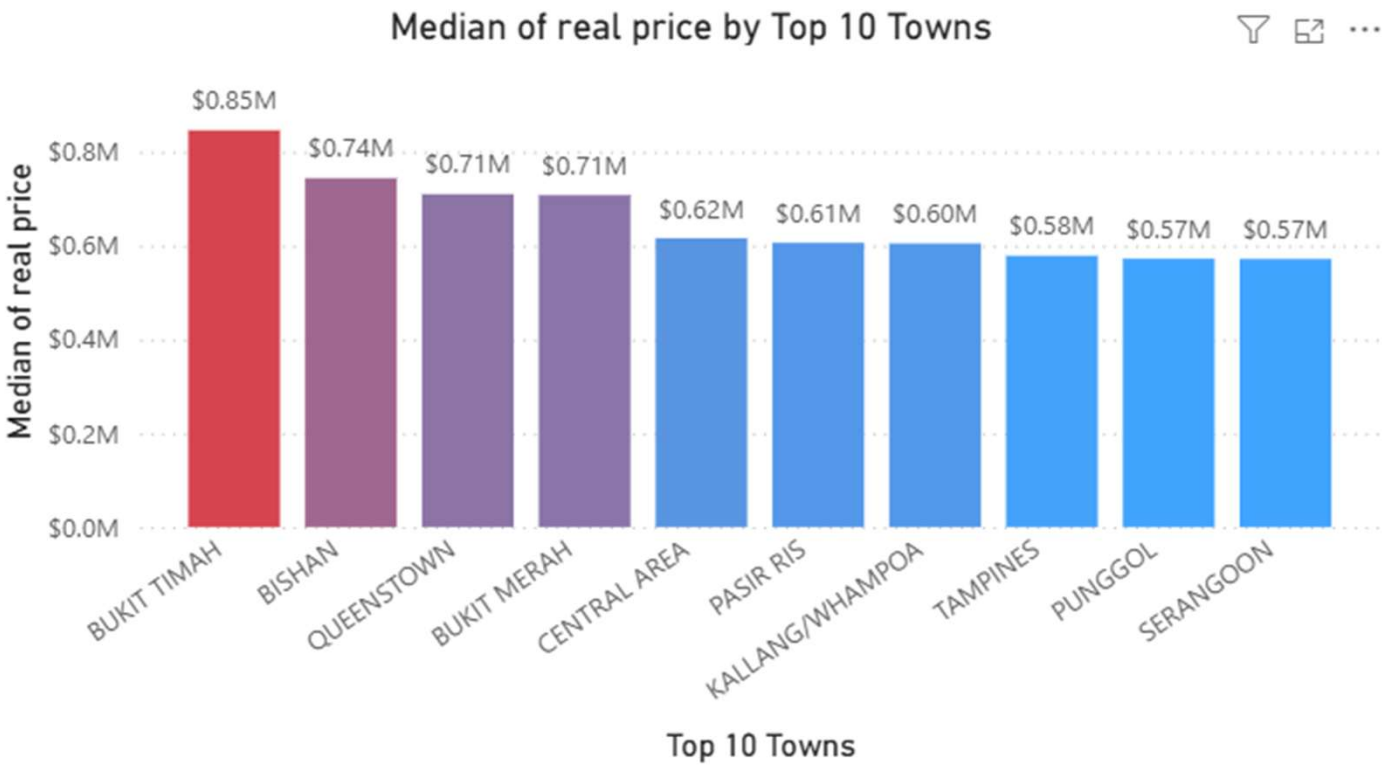
Lower floors, higher transactions. Higher floors, more premium.



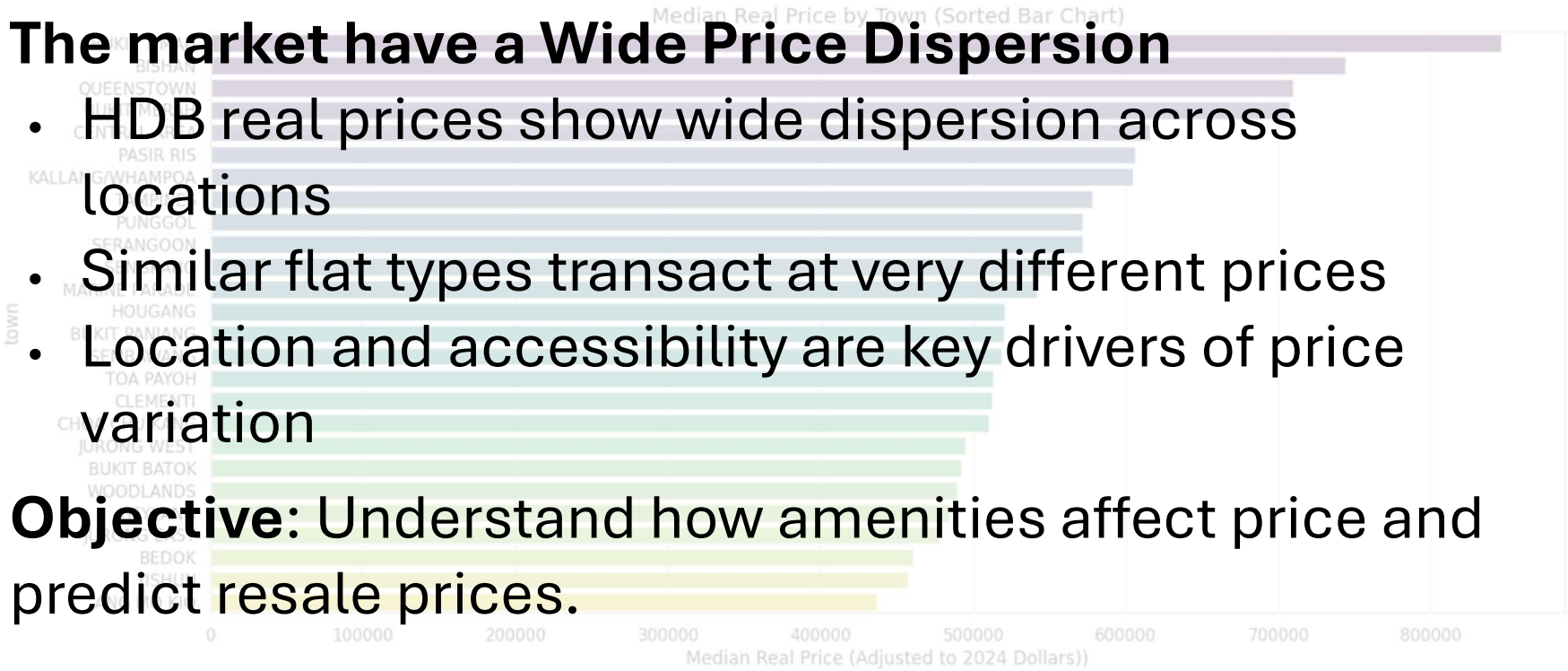


Older Flats tend to sell cheaper

Does location drive prices?



Location drive prices



The market have a Wide Price Dispersion

- HDB real prices show wide dispersion across locations
- Similar flat types transact at very different prices
- Location and accessibility are key drivers of price variation

Objective: Understand how amenities affect price and predict resale prices.

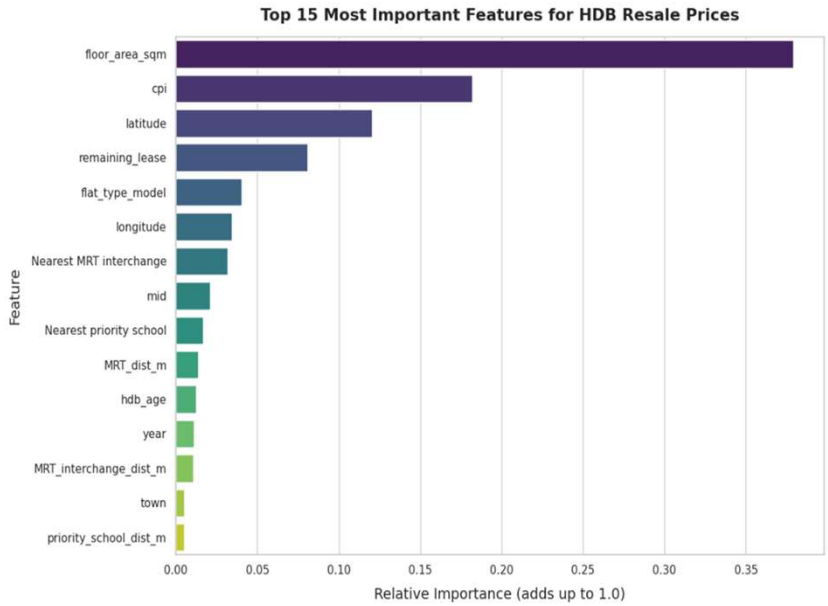
Price Difference drivers

- **Attributes That Influence Price**
  - Larger floor area correlates with higher resale prices
  - Higher storey units generally transact at a premium
  - Newer flats outperform older stock
- **Insight**

Size and age remain core value drivers.

- **Next Step**

Look at amenities like MRT and School



Impact of Schools on HDB Resale Prices

Is there a premium for flats nearer to Schools?

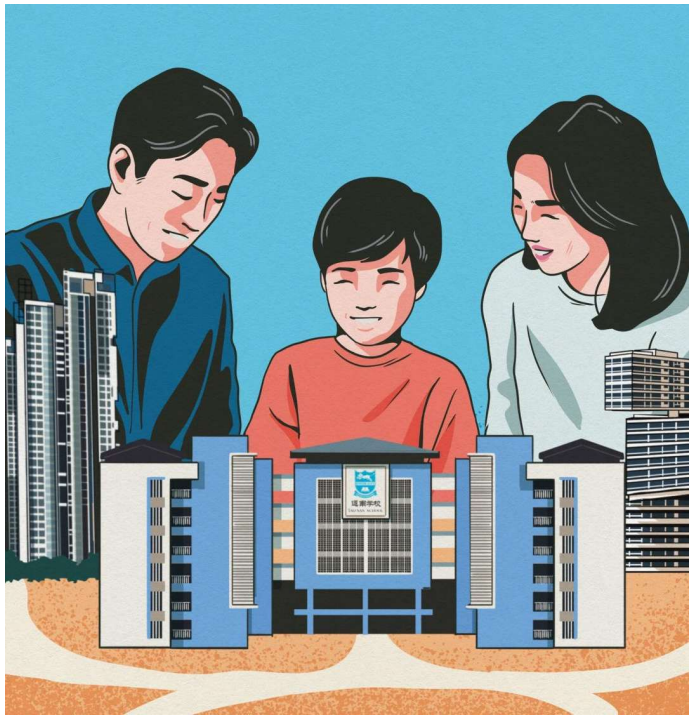
Primary school Priority & Policy

- **The Priority Hierarchy:** Enrollment is not a lottery; it is a phased competition where proximity is the "Tie-Breaker."
  - Phase 1 & 2A (The Insiders):

    Reserved for siblings, alumni, and staff. Proximity matters less here as spots are often guaranteed or highly prioritized.
  - Phase 2B (The Volunteers):

    For community leaders and school volunteers. 20 spots are reserved, and competition is fierce. Priority goes to 1km/2km.
  - Phase 2C (The Open Battle):

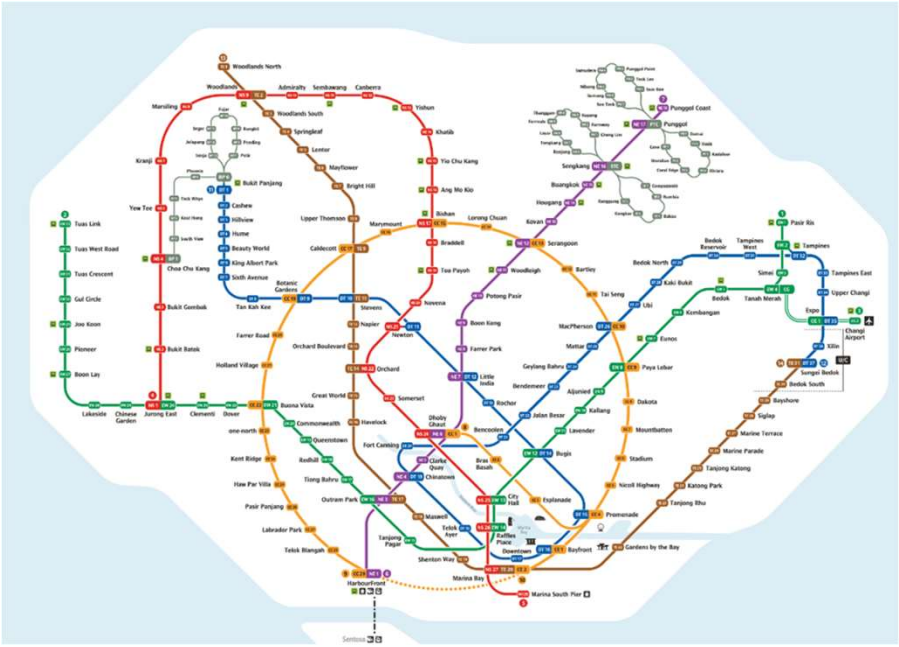
    This is where Residential Distance (1km/2km) becomes the deciding factor for the public.
- **The "1km Cliff":** Because Phase 2C prioritizes those within 1km, the demand for housing within this specific radius creates a "micro-market" price surge.
- **The 2027 Shift:** The current Gifted Education Programme (GEP) will be refreshed. I currently tracks GEP/SAP schools as they are the primary drivers of historical price surges.



The 1km School Entry Fee

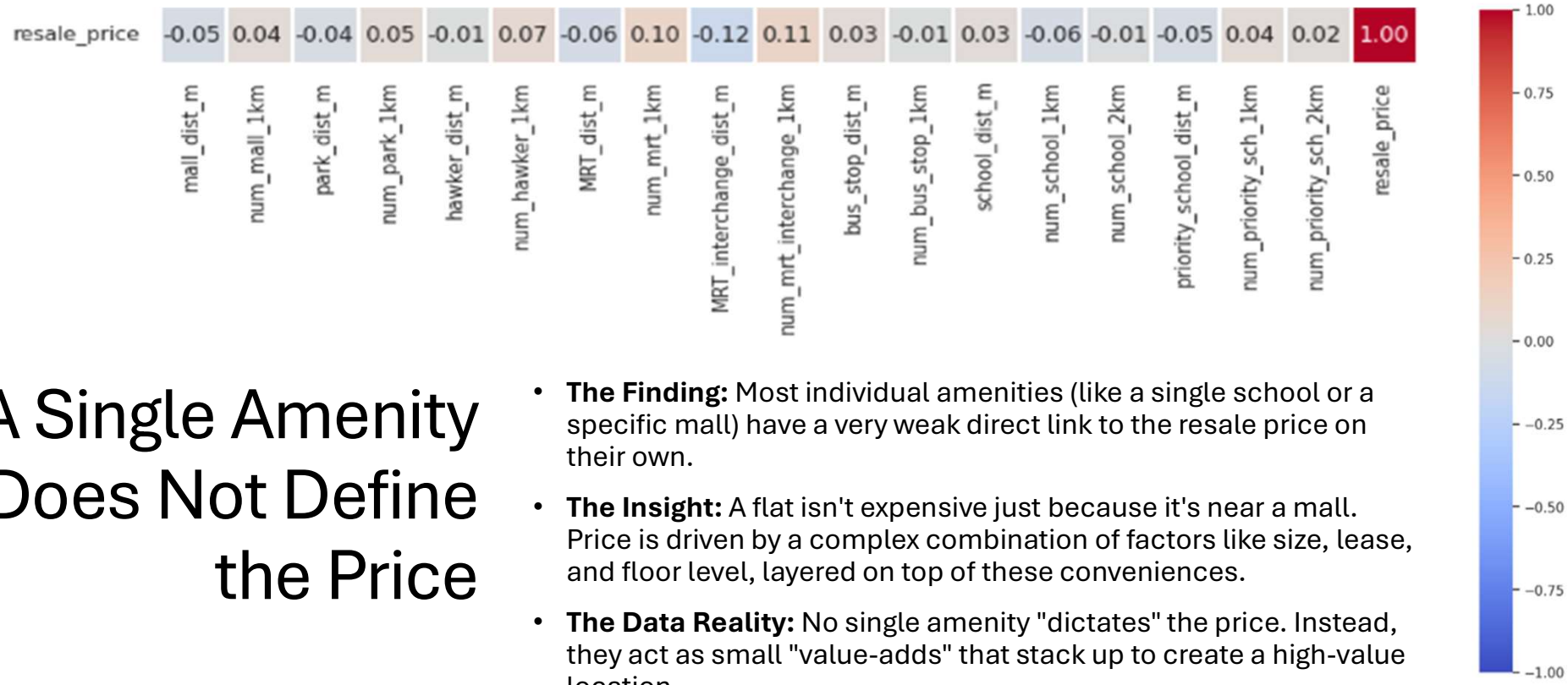
Flats within the 1km radius of GEP/SAP schools have a "surge" in price compared to flats just outside that line.

Buyers aren't just buying a home, they are buying "Priority Enrollment”, and the market prices it accordingly.



Impact of MRT stations on HDB Resale Prices

Is there a premium for flats nearer to MRT stations?



A Single Amenity Does Not Define the Price

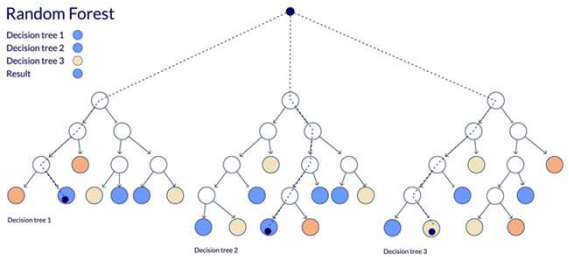
- **The Finding:** Most individual amenities (like a single school or a specific mall) have a very weak direct link to the resale price on their own.
- **The Insight:** A flat isn't expensive just because it's near a mall. Price is driven by a complex combination of factors like size, lease, and floor level, layered on top of these conveniences.
- **The Data Reality:** No single amenity "dictates" the price. Instead, they act as small "value-adds" that stack up to create a high-value location.

Transit is King: Identifying the Strongest Influencers

- **The Value Add (Positive Drivers):**
  - **Transit Choice**: Having multiple MRT stations within 1km **(+0.10)** is the most consistent way to boost a flat's value.
  - **Major Hubs**: Proximity to MRT Interchanges (where lines meet) **(+0.11)**has a stronger impact on price than a regular station.
  - **Food Convenience**: Being near more Hawker Centres **(+0.07)** provides a minor but measurable "foodie" premium.
- **The "Distance" Penalty (Negative Drivers):**
  - **The Interchange Gap**: The further from a major transit hub, the more the price tends to slide **(-0.12)**. This is the strongest "distance-based" relationship in the data.

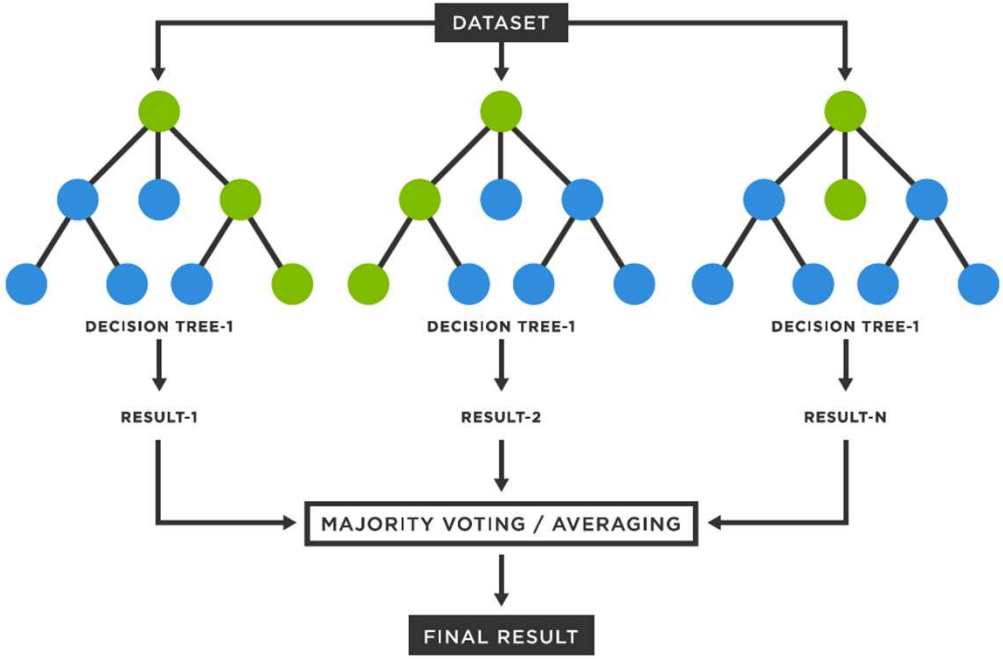
Why Some Amenities Don't Move the Needle Alone

- **The School & Park Paradox:** Being near a park or a top primary school shows almost no "straight-line" impact on price.
- **Why this happens:** These benefits are often localized. A school might only matter in a specific "hot" neighborhood, rather than raising prices everywhere in Singapore equally.
- **Modeling:** Because these individual links are so low, a "Linear regression" model won't work.
- **The Solution:** Random Forest will be a better model. It doesn't look for one single reason for a high price; it captures the "Lifestyle Value" by looking at how all these small factors work together.



Random Forest: A Team of Decision Trees

- A decision tree is like asking a series of yes/no questions.
- One tree can be wrong / biased.
- A random forest combines many trees, each looking at different parts of the data
- Together, they “vote” or “average” the best answer
- This teamwork makes predictions more accurate and less risky.



Recommendations for Buyers & Agents

- **For Buyers:** Use the predictive map to spot "overpriced" flats that don't justify their proximity premium.
- **For Agents:** Use the predictive map to show sellers why their house is valued the way it is compared to the neighbor's.

Town

All

Flat type model

All

Floor area sqm

31 367

Storey range

All

Lease commence date

All

1 BEACH RD

$489,236
Predicted Price

$7,023
Predicted price per sqm

$492,888
Historical Resale price

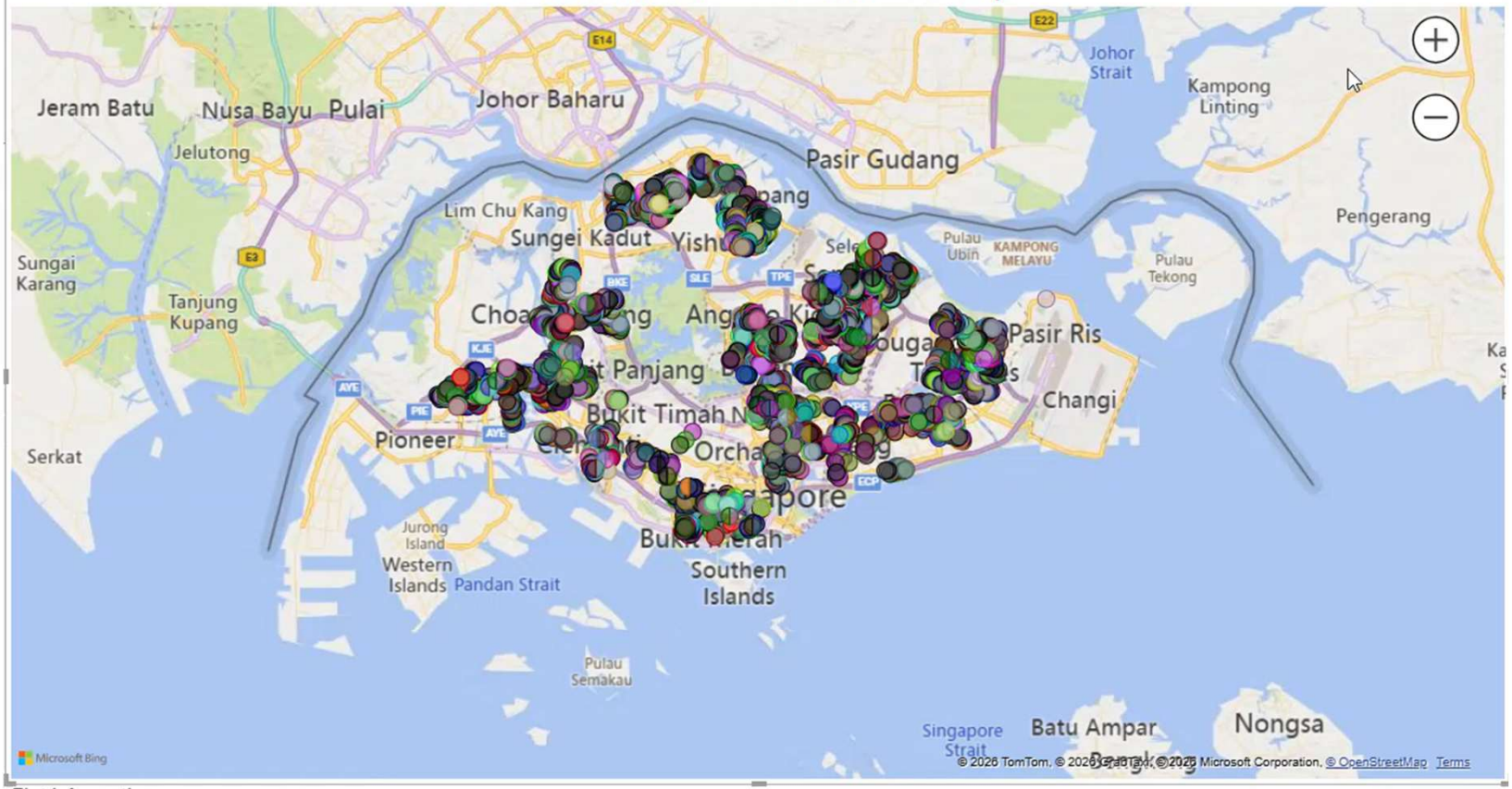
$7,248
Historical resale price ...

Admiralty
Nearest MRT Station

ADMIRALTY PRIMARY SCHOOL
Nearest school

AI TONG SCHOOL
Nearest top primary

HDB Resale flats Predictive Price Map



Flat information

ANG MO KIO	2 ROOM Improved	01 TO 03	44	48	7
Town	Flat type model	Storey range	Floor area sqm	Current HDB age	No. of Transactions

Amenities

17	1	0	1	4	1	2
Bus stop <1km	Hawker <1km	Mall <1km	MRT <1km	Park <1km	Top Primary <1km	School <1km

LIMITATIONS

Policy and interest rate factors not included

No variables for, **Government policy changes**, loan interest rates.
As a result, future price predictions may differ from historical trends, and the model cannot predict policy or economic shocks.

Feature limitations

Does not include qualitative features such as, renovation quality, interior condition view or orientation and unit-specific attributes. **These factors may affect resale price but are not captured.**

Assumption of stable relationships

Assumes historical relationships between features, resale price and government housing policies remain stable.

THE
END

Leveling the Playing Field: Adjusting for Inflation

The Challenge: A $400,000 flat in 2015 is not the same as a $400,000 flat today.

General inflation changes the value of the Singapore Dollar over time.

The Solution: "2024 Dollars“

Using the Consumer Price Index (CPI) from SingStat to convert every historical transaction into a consistent "Base Year" value (2024).

✓ **How it Works**:

- A flat sold in 2015 for $400k is adjusted up to reflect what that money is worth today.
- A flat sold in 2025 for $600k is adjusted down to remove the "noise" of general price hikes.

✓ **The Benefit**:

- Ignore general inflation
- Focus purely on real property trends

Year

All

Town

All

Flat type

All

Storey range

All

Lease commence date

1/1/1966 1/1/2021

Floor area sqm

31 367

YISHUN

260019

No. of Transactions

93

Floor area sqm

$478,000

Resale price

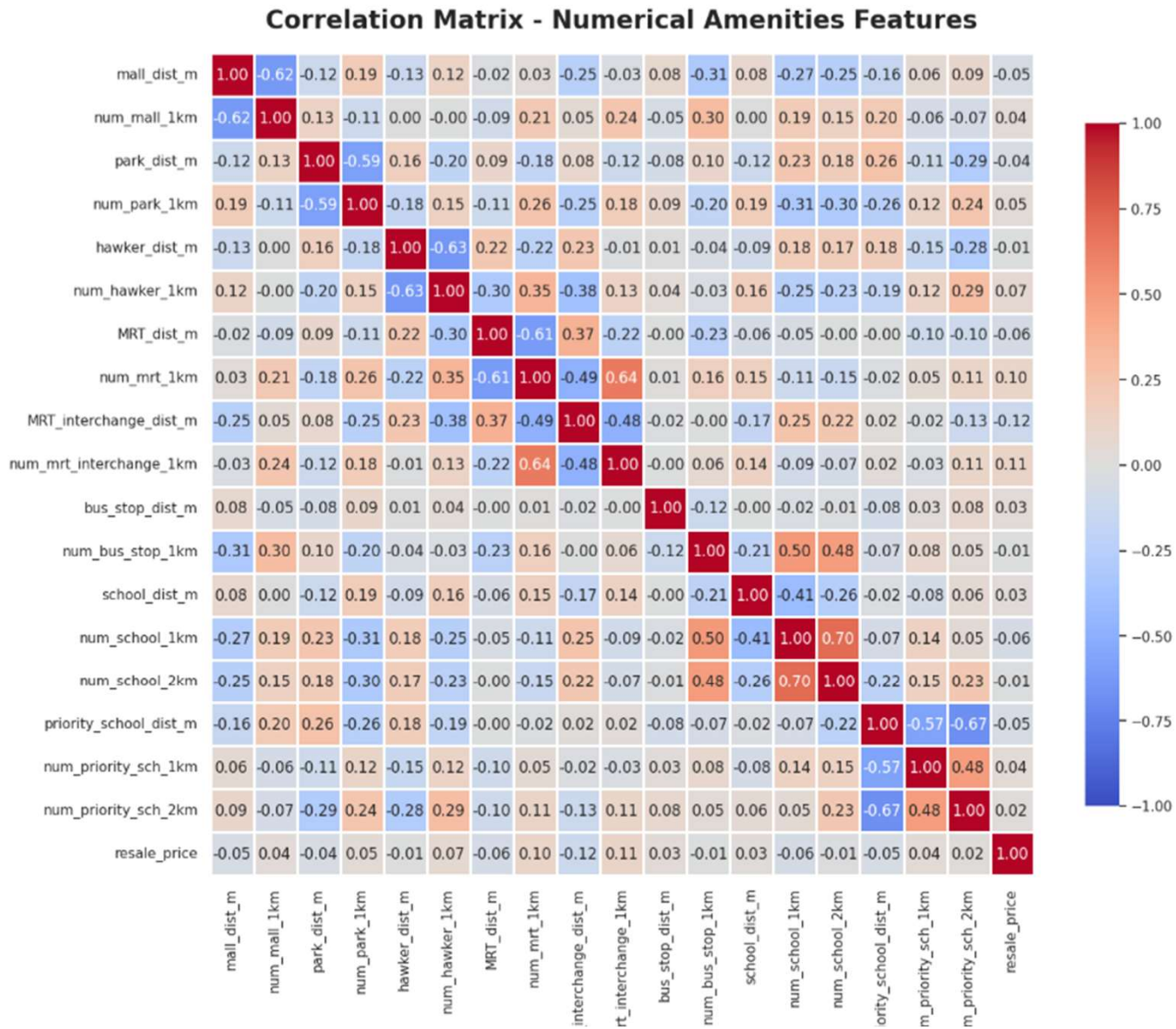
$5,049.18

Price per sqm

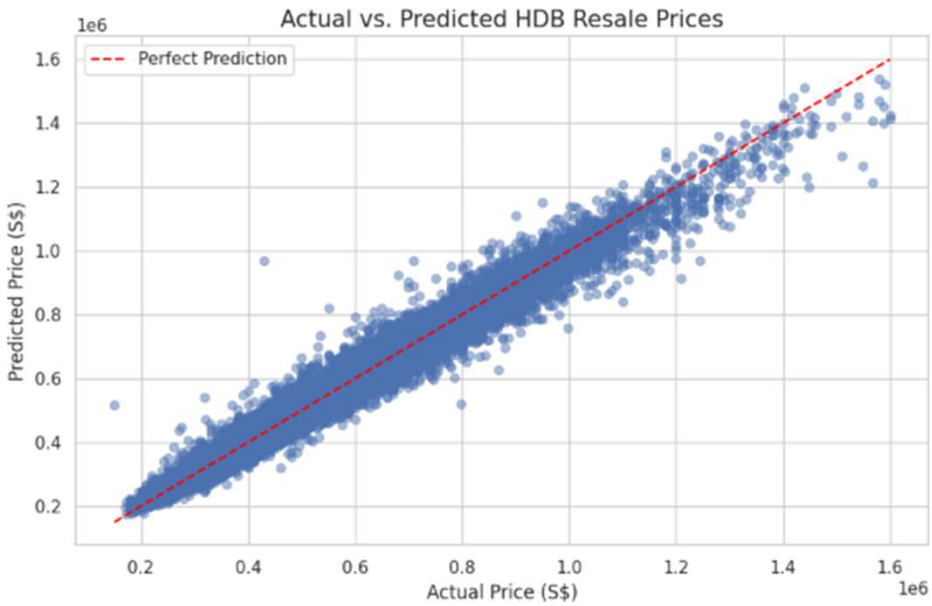
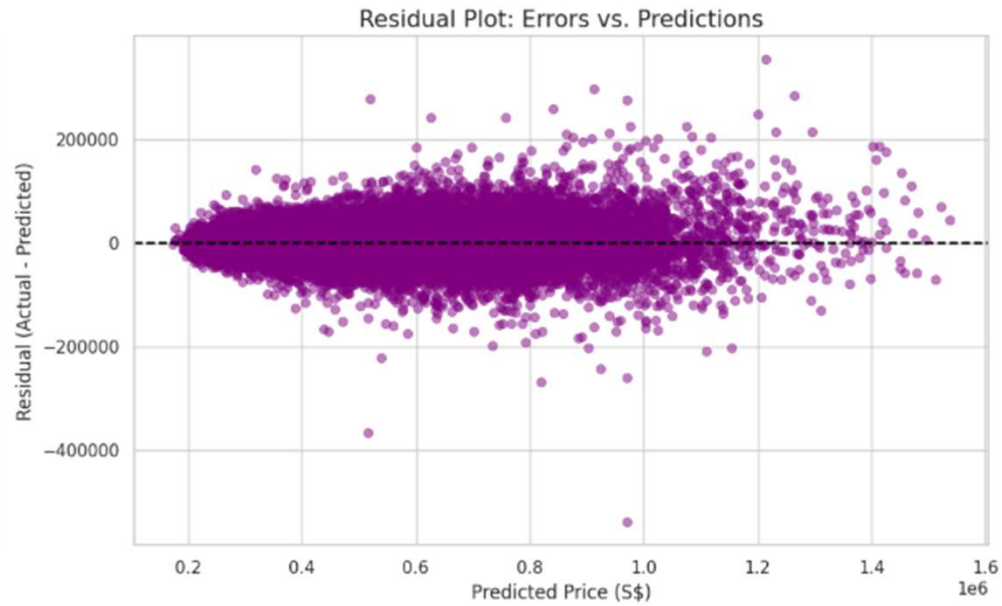
No. of Transactions by Town

Median of real price by Town

A Single Amenity Does Not Define the Price



My Model Results?

- It is **highly reliable** for predicting a "Fair Value" for a flat.
- **R^2 = 0.9773**

Model	Train R2	Test R2	Test RMSE	Test MAE
Dummy Regressor (Baseline)	0.0000	-0.0000	$183974	$144202
Linear Regression	0.8987	0.8972	$58996	$44157
Linear Regression (Log)	0.9267	0.9112	$54810	$36676
Ridge Regression	0.8987	0.8971	$59001	$44164
Random Forest	0.9968	0.9773	$27693	$19219